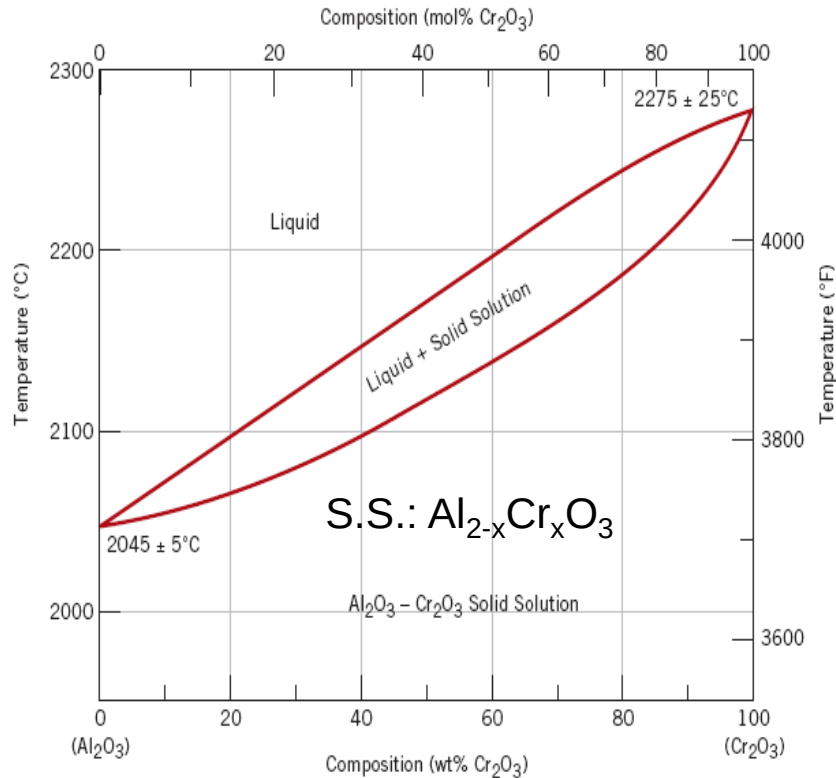


TOPIC 4. Phase Diagrams IV

1. Ceramic phase diagrams
2. Ternary phase diagrams
3. Ternary phase diagrams in metallic alloys
4. Ternary phase diagrams in ceramics

CERAMIC PHASE DIAGRAMS



- Temp $< 950^{\circ}\text{C}$: immiscibility: existence of two crystalline phases or solid solutions

- Complete interval of solid solution $\text{Al}_{2-x}\text{Cr}_x\text{O}_3$ below the solidus temperature ($\sim 2100^{\circ}\text{C}$) and 950°C .

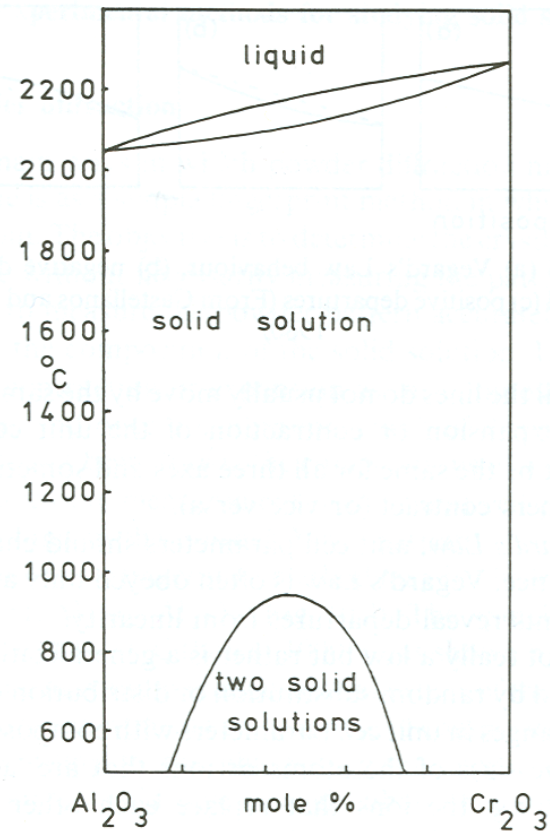
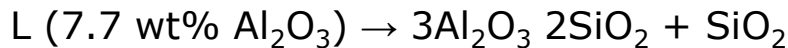


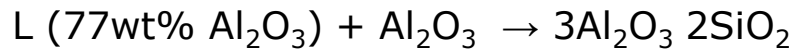
Fig. 10.6 Phase diagram for Al_2O_3 - Cr_2O_3 . (Data from Bunting, 1931, and Roy and Barks, 1972)

CERAMIC PHASE DIAGRAMS

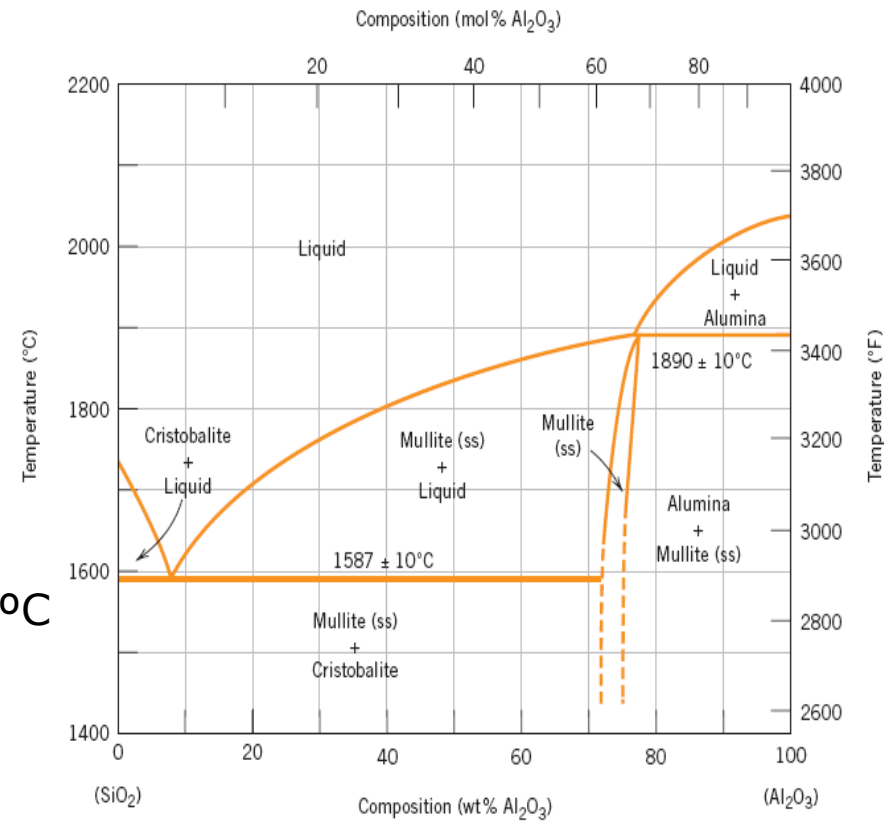
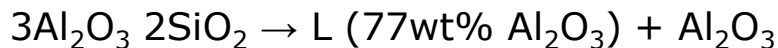
Invariant eutectic reaction at 1587°C



Invariant peritectic reaction at 1828°C



S.S. Mullite: Incongruent melt. point at 1828°C

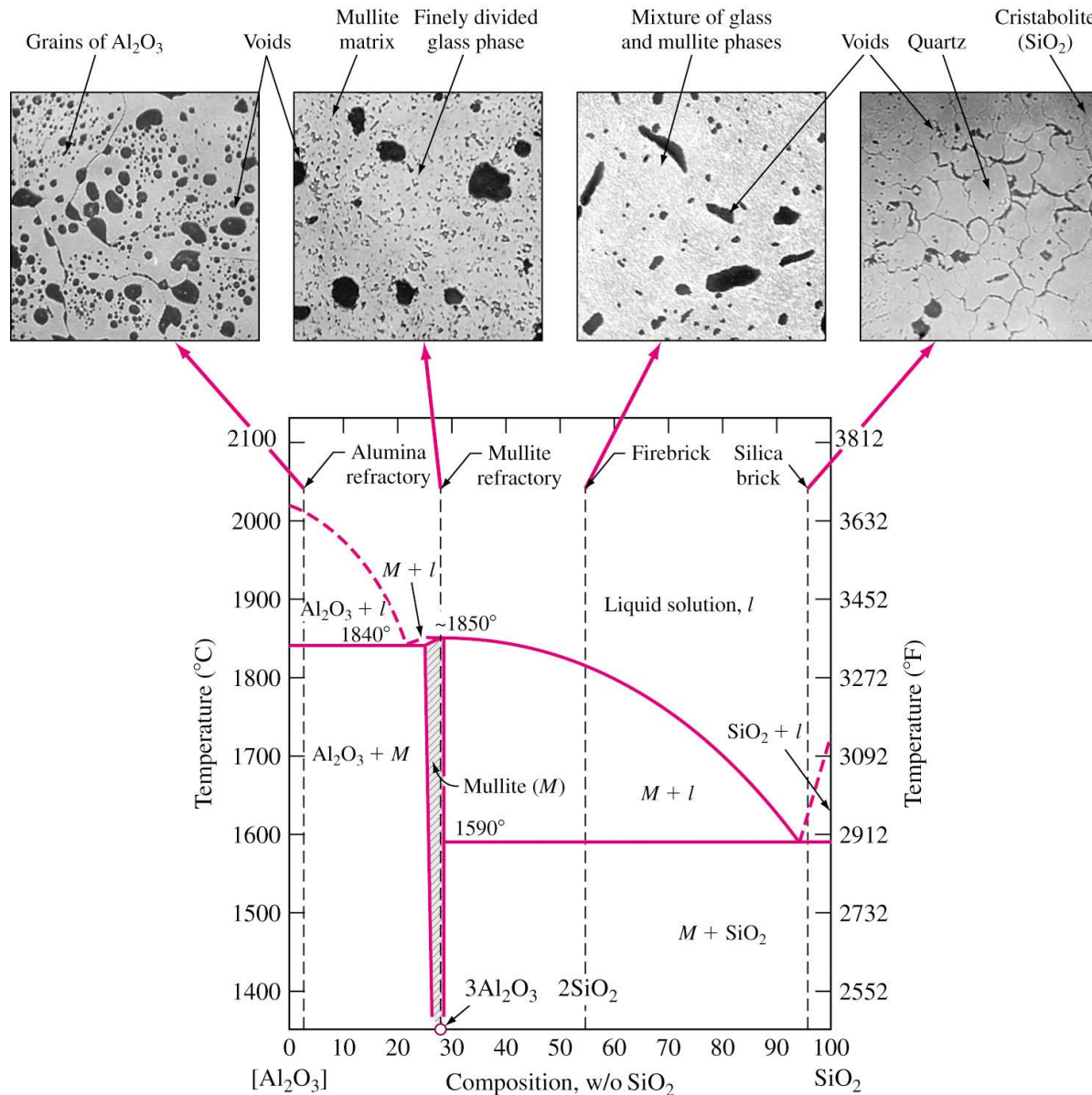


Mullite: $3\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$

Alumina: Al_2O_3

Cristoballite: SiO_2

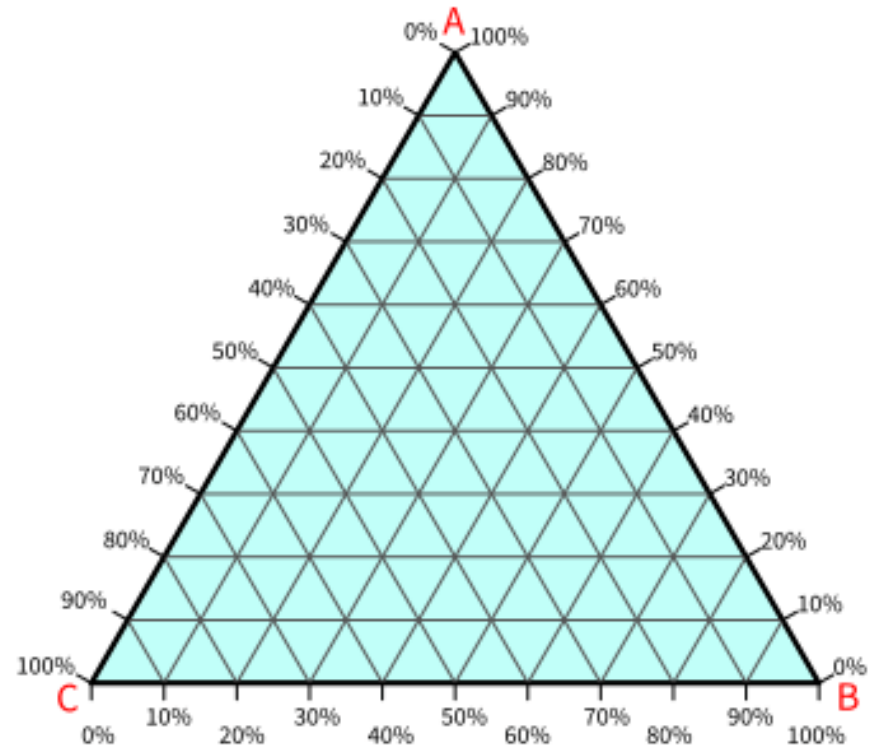
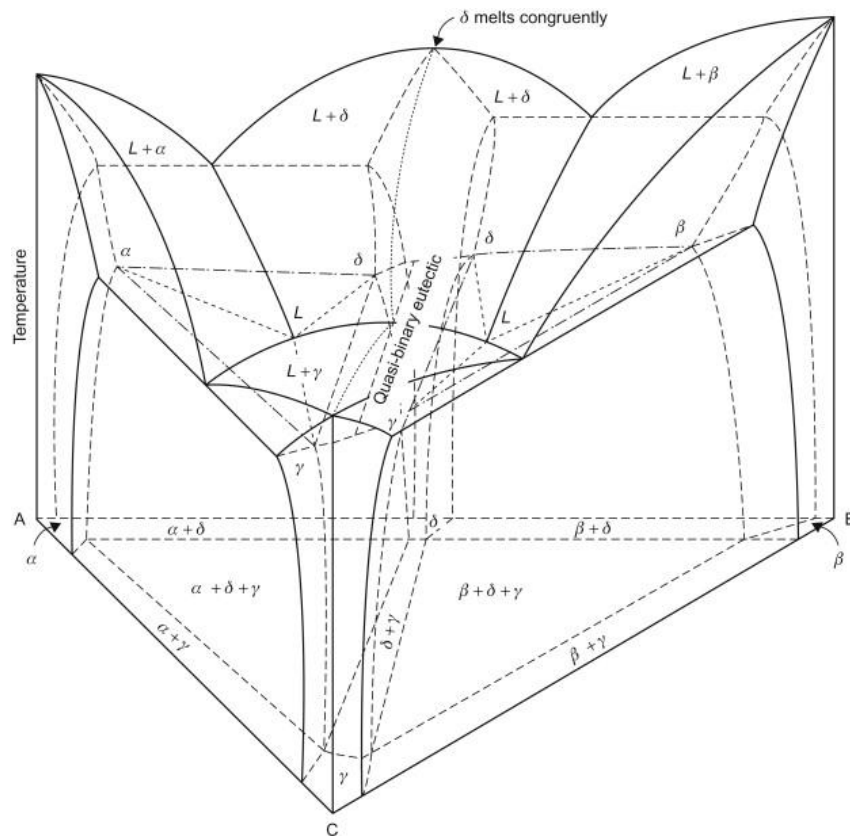
Intermediate Phases in Ceramics



In Al_2O_3 - SiO_2 system, an intermediate phase called Mullite is formed, which includes the compound $3\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$.

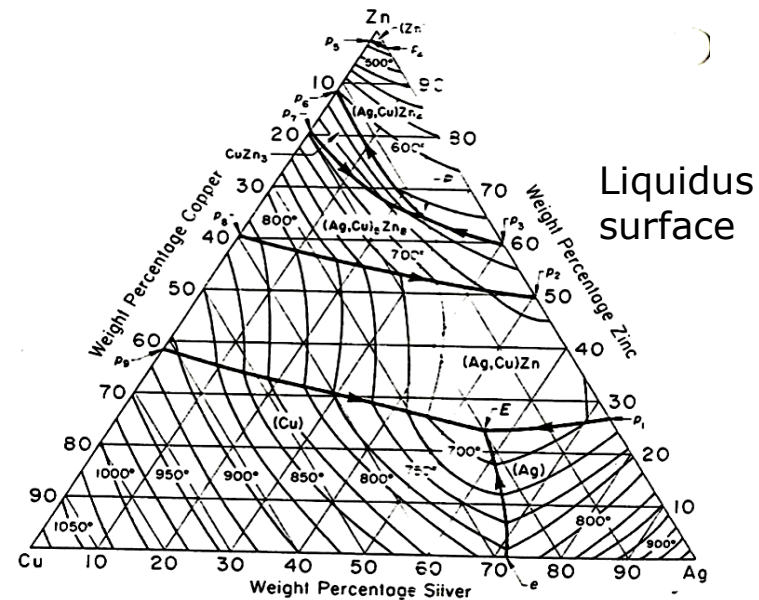
TERNARY PHASE DIAGRAMS

- **3 components** (A, B, C)
- **Corners:** Pure components
- The **sides** represent **binary compositions** A-B, A-C or B-C

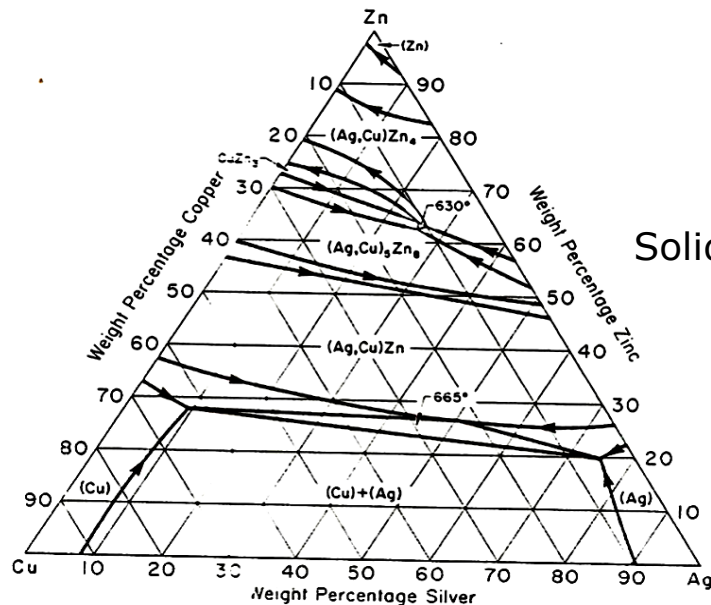


TERNARY PHASE DIAGRAMS IN METALLIC ALLOYS

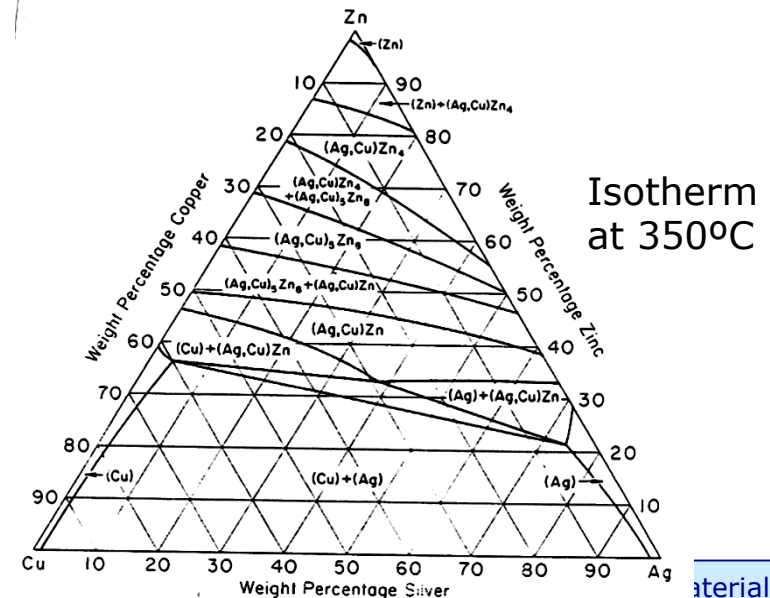
Ag-Cu-Zn System



Liquidus surface



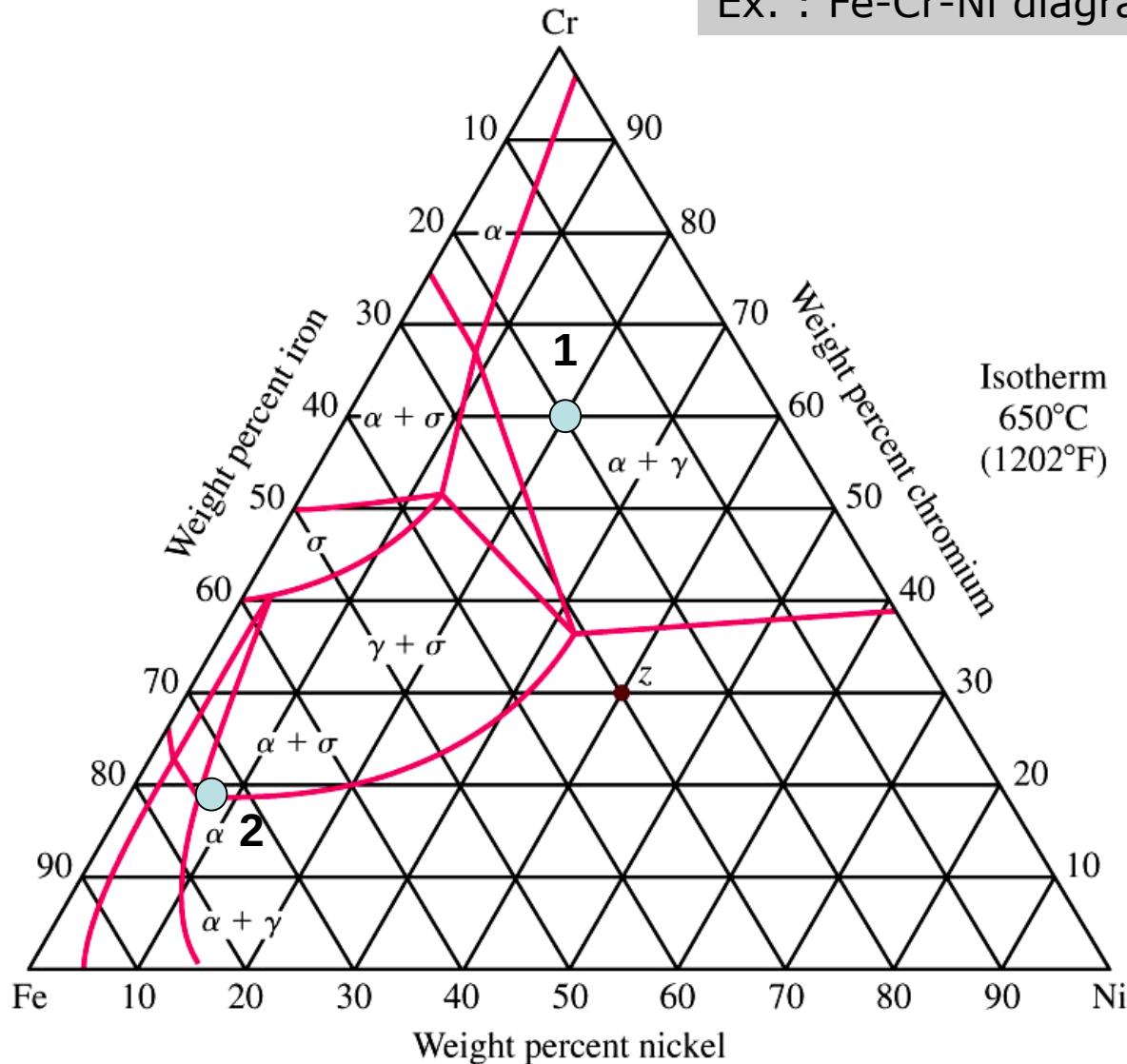
Solidus surface



Isotherm at 350°C

TERNARY PHASE DIAGRAMS

Ex. : Fe-Cr-Ni diagram (stainless steel)



Point 1:

Phases $\alpha + \gamma$

20%Fe

60%Cr

20%Ni

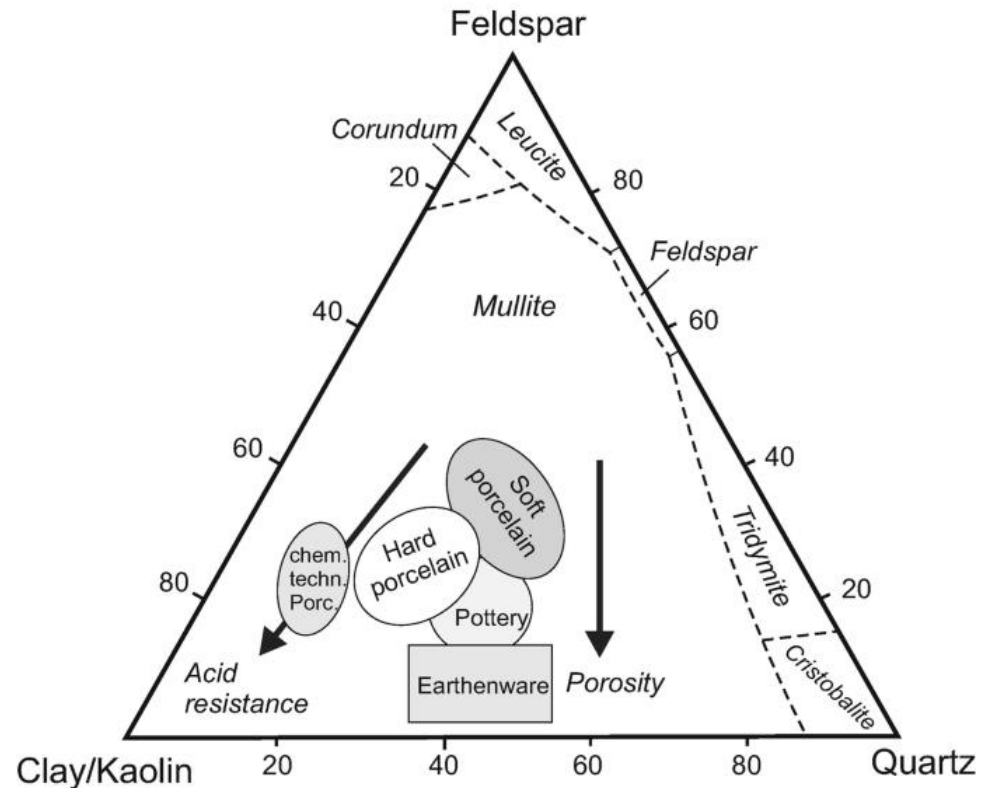
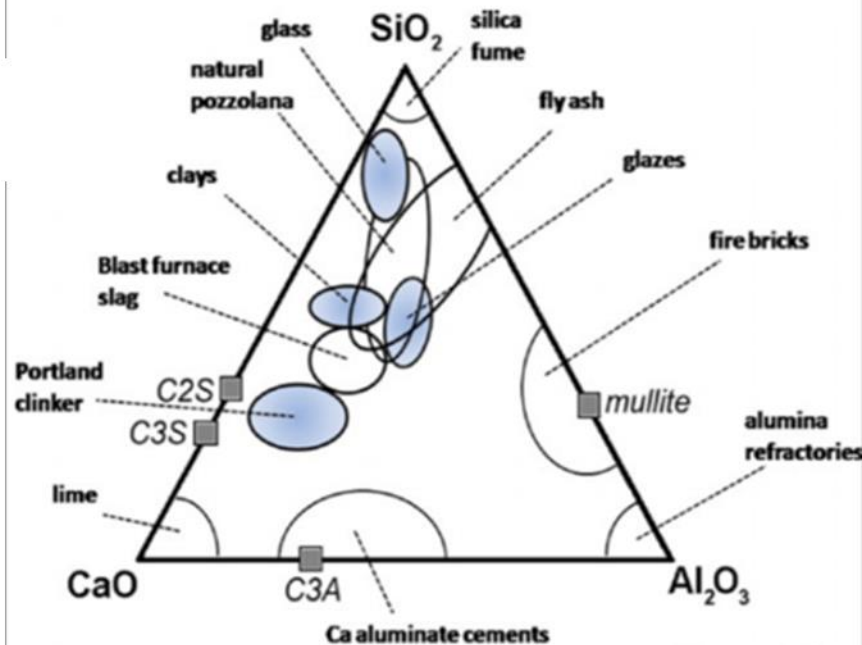
Isotherm
650°C
(1202°F)

Point 2:

Commercial stainless
steel: 74%Fe + 18%Cr +
8%Ni

CERAMIC TERNARY PHASE DIAGRAMS

Ex : Ceramic diagrams



Composition of traditional ceramics:
clay-feldspar-quartz